and calculation of power and energy in electric circuit.

i) Work

Work is said to be done by Force 'F' when the point of its application moves through a distance 'S'.

Mathematically, Work = Force × distance = F × S =FS.

The unit of force is Newton (N). If 1 Newton force moves a body to a distance of 1 metre, then the work done is 1Nm (Newton - metre)

In an electric circuit, if 1 volt electric potential causes 1 coulomb of electric charge to pass through it, then the work done is equal to 1 joule.

1 joule = 1 volt × 1 coulomb

Coulomb = Ampere × time

i.e $J = V \times I \times t$

ii) Power

Power is the rate of doing work. Its unit is watt (W).

$$\text{Power} = \frac{\text{work done}}{\text{time}} = \frac{\text{Joule}}{\text{time}} = \frac{V \times I \times t}{t}$$

$$\text{Power } P = VI \text{ watt } [V = IR,\ P = I^2R]$$

1 KW = 1000 watt

1 HP = 746 watt

iii) Energy

The amount of work done by an equipment during a time period of 't' seconds. The unit of energy is joule.

Energy = power × time watt sec

The energy spent for the appliances in 1 kwh is called as one unit i.e

1 unit = 1kWh

The power of iron box is 1000 watt. when used for 1 hour, the energy consumed is, 1000 watts × 1 hour = 1000Wh = 1kWh = 1unit.

Example: 1

The resistance of a lamp is 10Ω and 2A current is flowing through it. Calculate the power of the lamp.

Solution:

Resistance (R) = 10Ω

Current (I) = 2A

Power = $I^2.R$

$=2^2 \times 10 = 40W$

Example: 2

In a factory, the following appliances are used

a) 3hp motor works 5 hours per day.
b) 100W capacity of 40 lamps glow 8 hours per day.
c) 1500W capacity of heater works 6 hours per day.

Calculate the cost of energy consumed in 30 days. (1unit cost is Rs. 6.00)

Solution:

a) 3hp motor works 5 hours per day

$$\text{Energy} = \frac{3 \times 746 \times 5}{1000} = \frac{11190}{1000} \text{Wh}$$

$$= 11.190 \text{ kwh} = 11.190 \text{ unit}$$

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